

When to screen?

5 domains of
prevention/screening

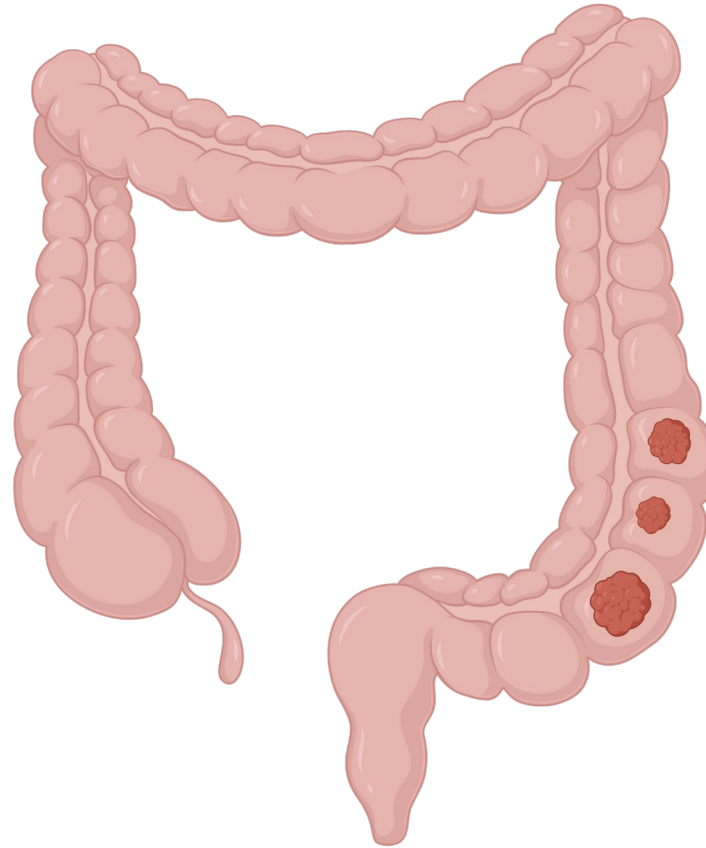
Cancer Screening

Infectious Disease

CASES

Adult Disease Prevention

Part 1 – Cancer Screening and Immunizations



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Conditions for Effective Screening

**DISEASE
CHARACTERISTICS?**

Common
Asymptomatic period
Early > Late treatment

**SCREENING TEST
CHARACTERISTICS?**

Acceptable sensitivity/specificity
Low-cost
Easy to complete
Minimal patient harm



BOTTOM LINE:

BENEFITS > RISKS/HARMS

When to screen?

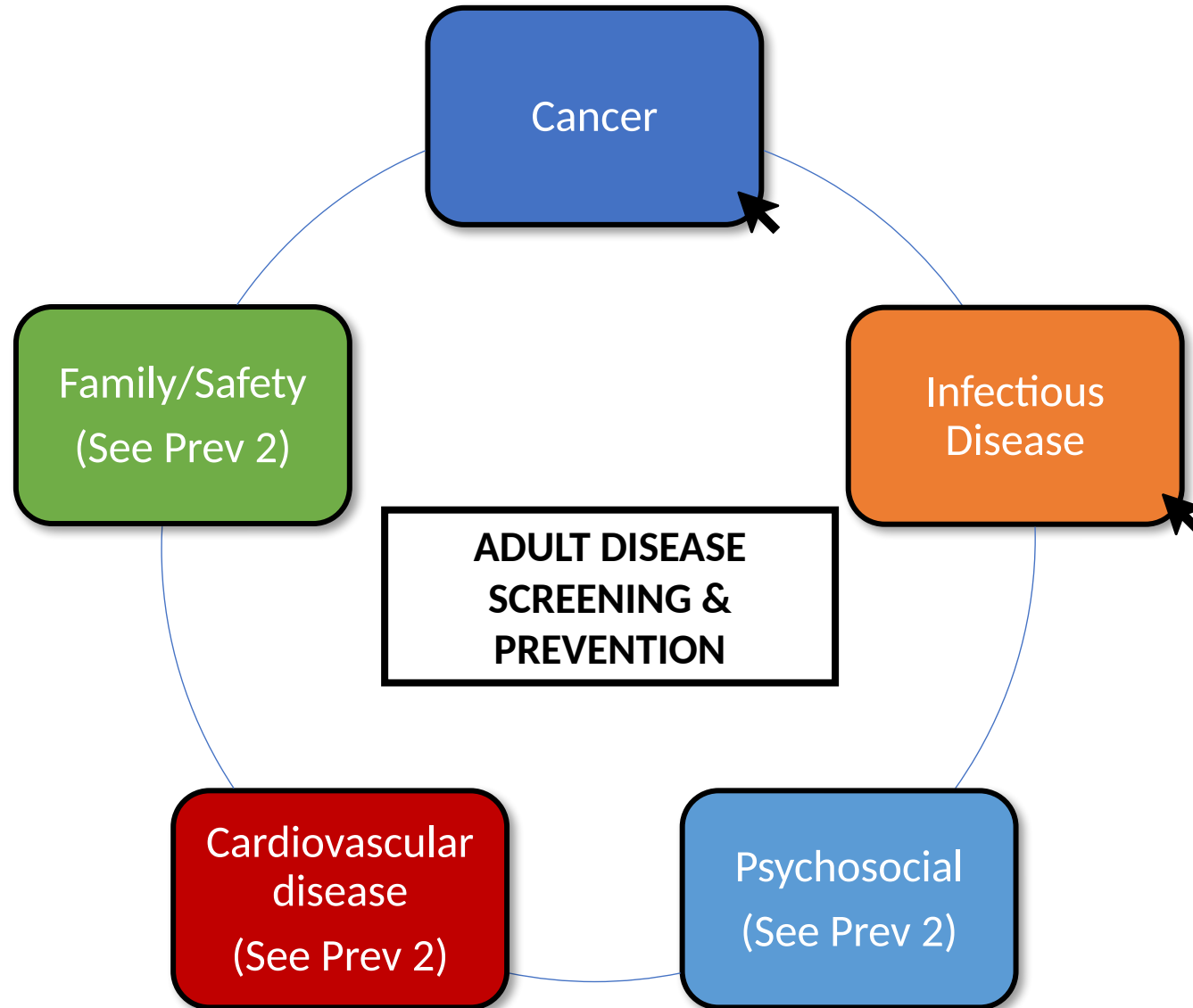
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Cancer Screening

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Adult Disease Prevention – Average Risk Patient

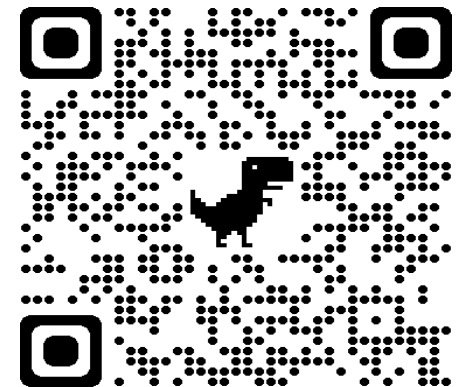


Prioritize Screening
and Prevention
Based On:

1. Age
2. Sex
3. Risk Factors

Can't keep it straight?

Use the USPSTF Tool



Breast Cancer Screening

When to screen?

5 domains of prevention/screening

Cancer Screening

Breast

Colon

Cervical

Lung

Prostate

Infectious Disease

CASES

Population?

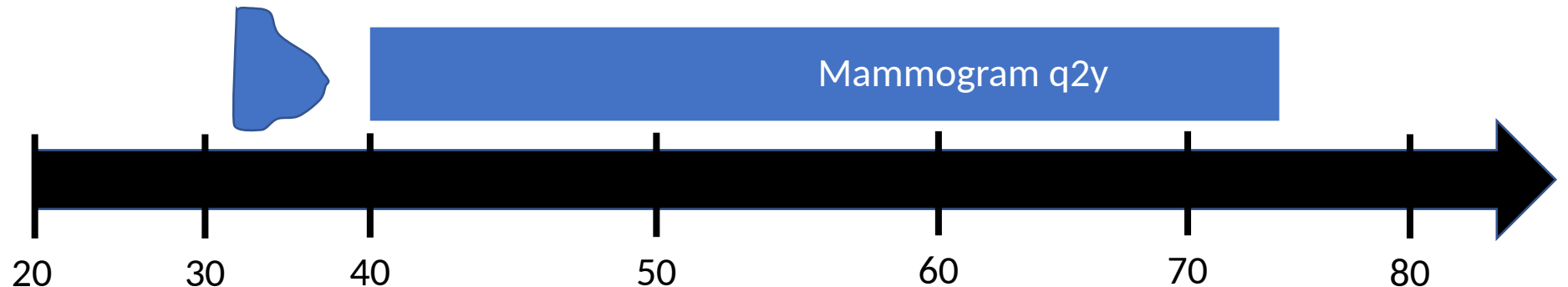
All average-risk females ages 40-74 (Grade B)

Modality and Interval?

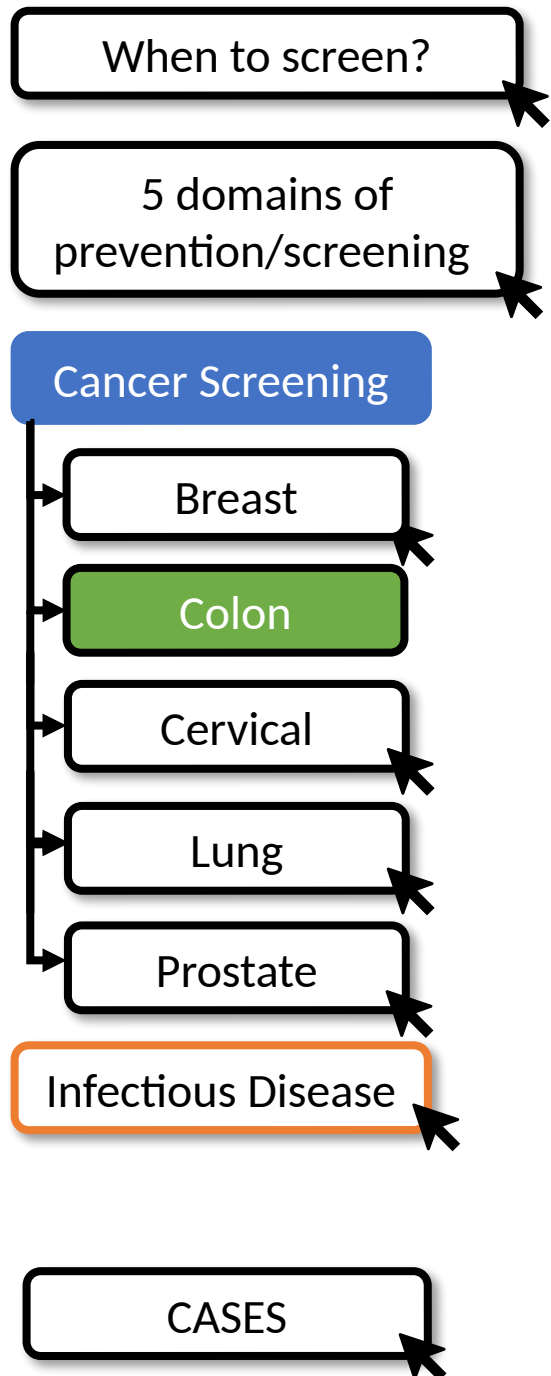
Mammogram every 2y

Discontinuation?

If life expectancy is <10y
OR
>74



Colon Cancer Screening



Population?

Average-risk patients ages 45-79
(age 45-49 is Grade B)
(age 76-84 is Grade C)

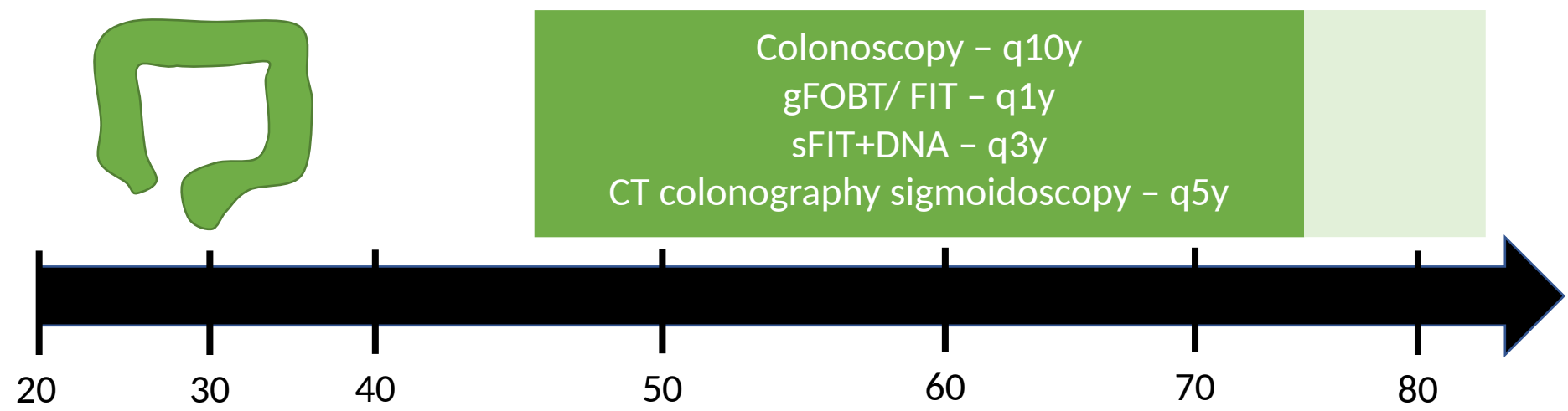


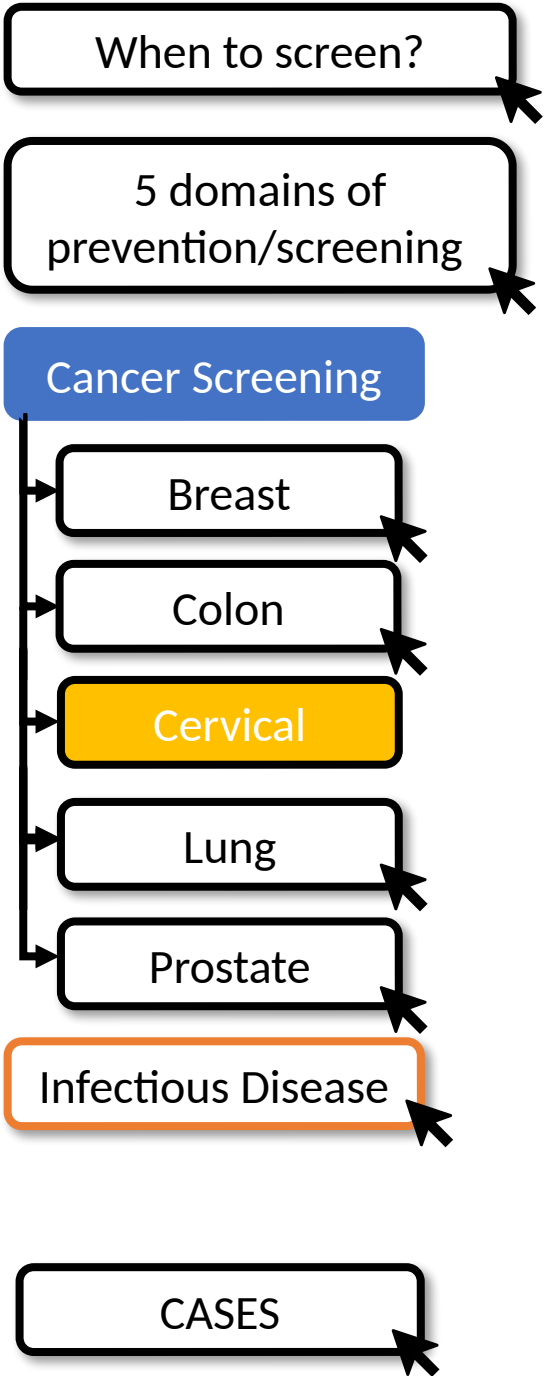
Modality and Interval?

Colonoscopy	q10y
gFOBT/FIT	q1y
sFIT+DNA	q3y
CT colonography or sigmoidoscopy	q5y

Discontinuation?

Selectively offer in patients age 75 to 85 based on patient factors.





Cervical Cancer Screening

Population?

USPSTF Guidelines
Starting age 21 (Grade A)

ACS Guidelines
Starting age 25

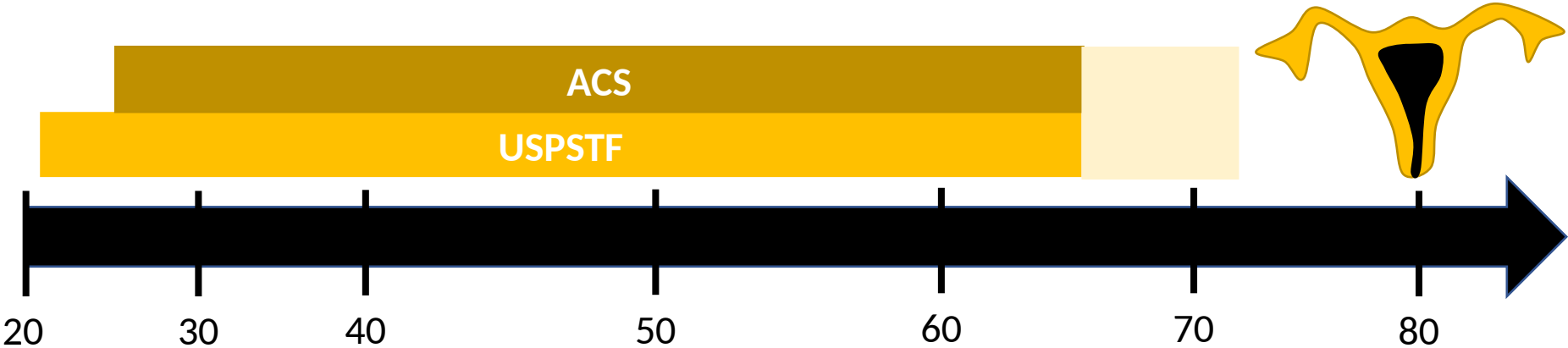
Modality and Interval?

21-29y	Cytology q3y
30-65y	Cytology q3y HPV ± cytology q5y

Primary HPV	q5y
HPV + cytology	q5y
Cytology	q3y

Discontinuation?

>65y with normal screening in prior 10y
OR
Hysterectomy with removal of cervix + no prior history of high-grade precancerous lesion



Lung Cancer

When to screen?

5 domains of prevention/screening

Cancer Screening

Breast

Colon

Cervical

Lung

Prostate

Infectious Disease

CASES

Population?

Ages 50-80 with at least
20 pack-year history +
smoked within the past 15y
(Grade B)

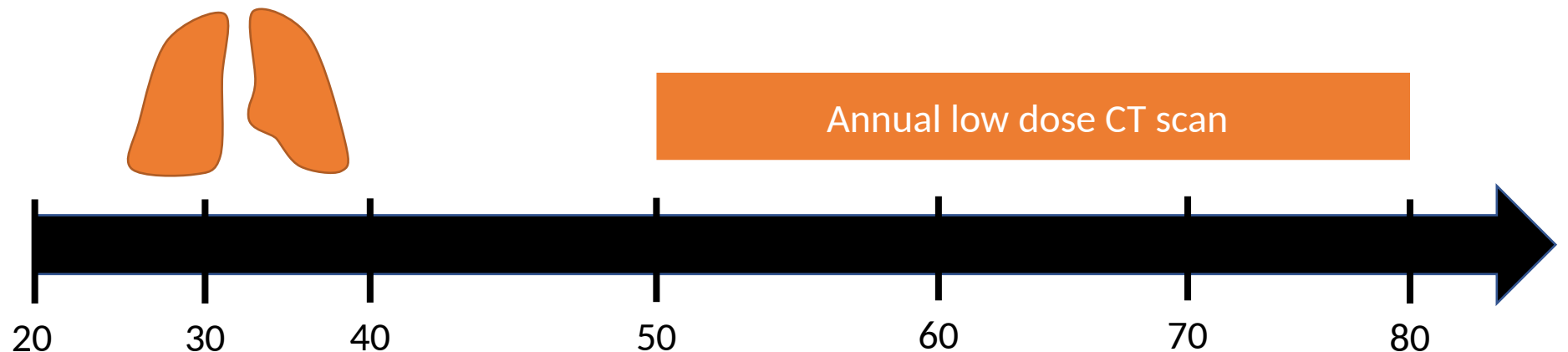
Modality and Interval?

Annual low-dose CT scan

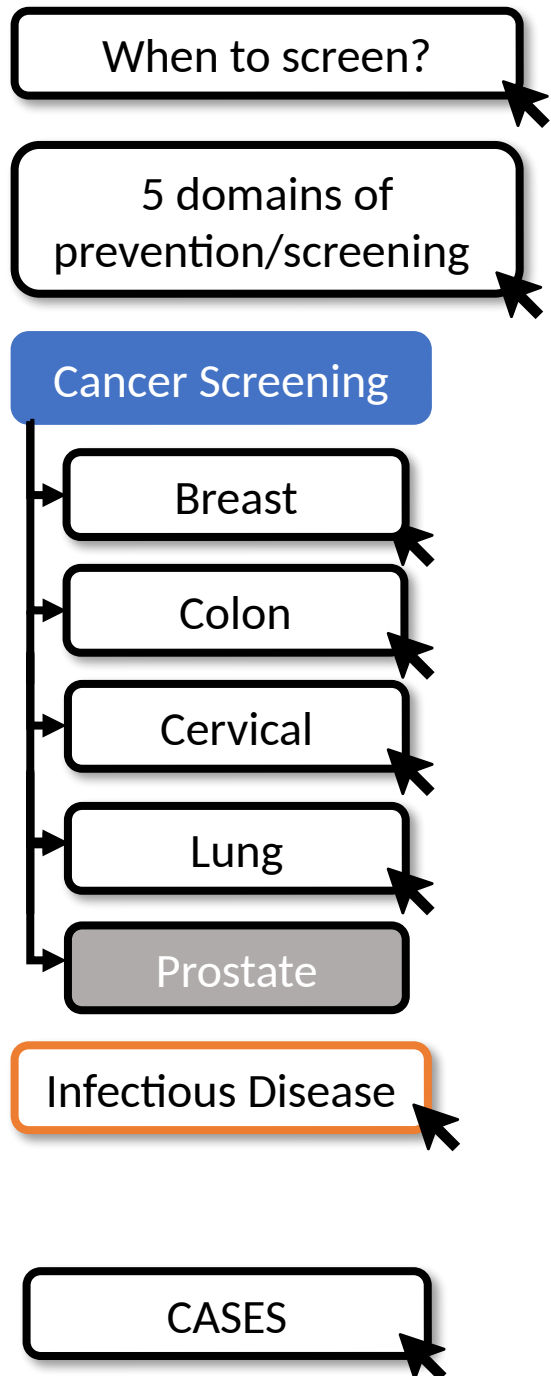
Discontinuation?

If quit smoking >15y ago
OR
Limited life expectancy

Medicare requires a **shared decision-making** visit prior to first screening.



Prostate Cancer Screening



Population?

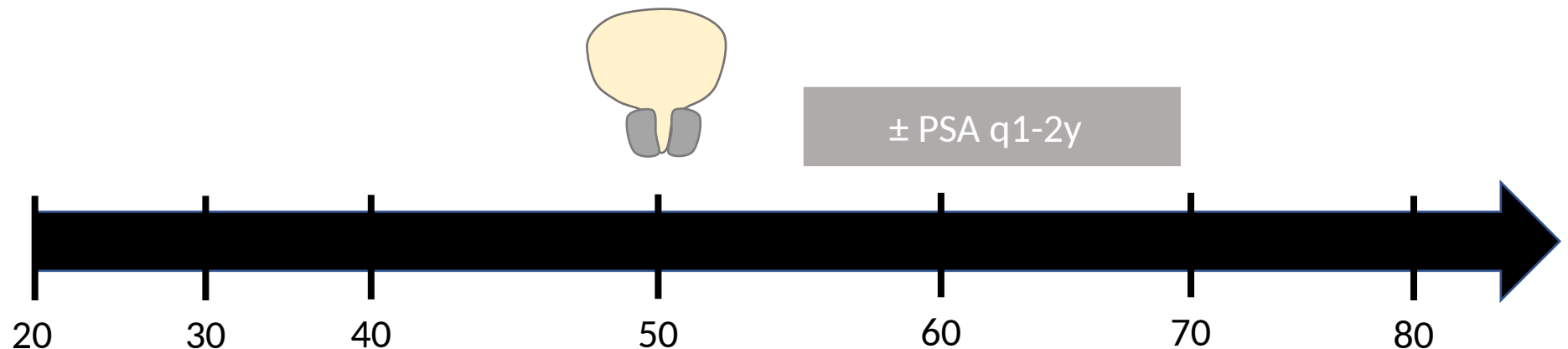
Males aged 55-69y
(Grade C)

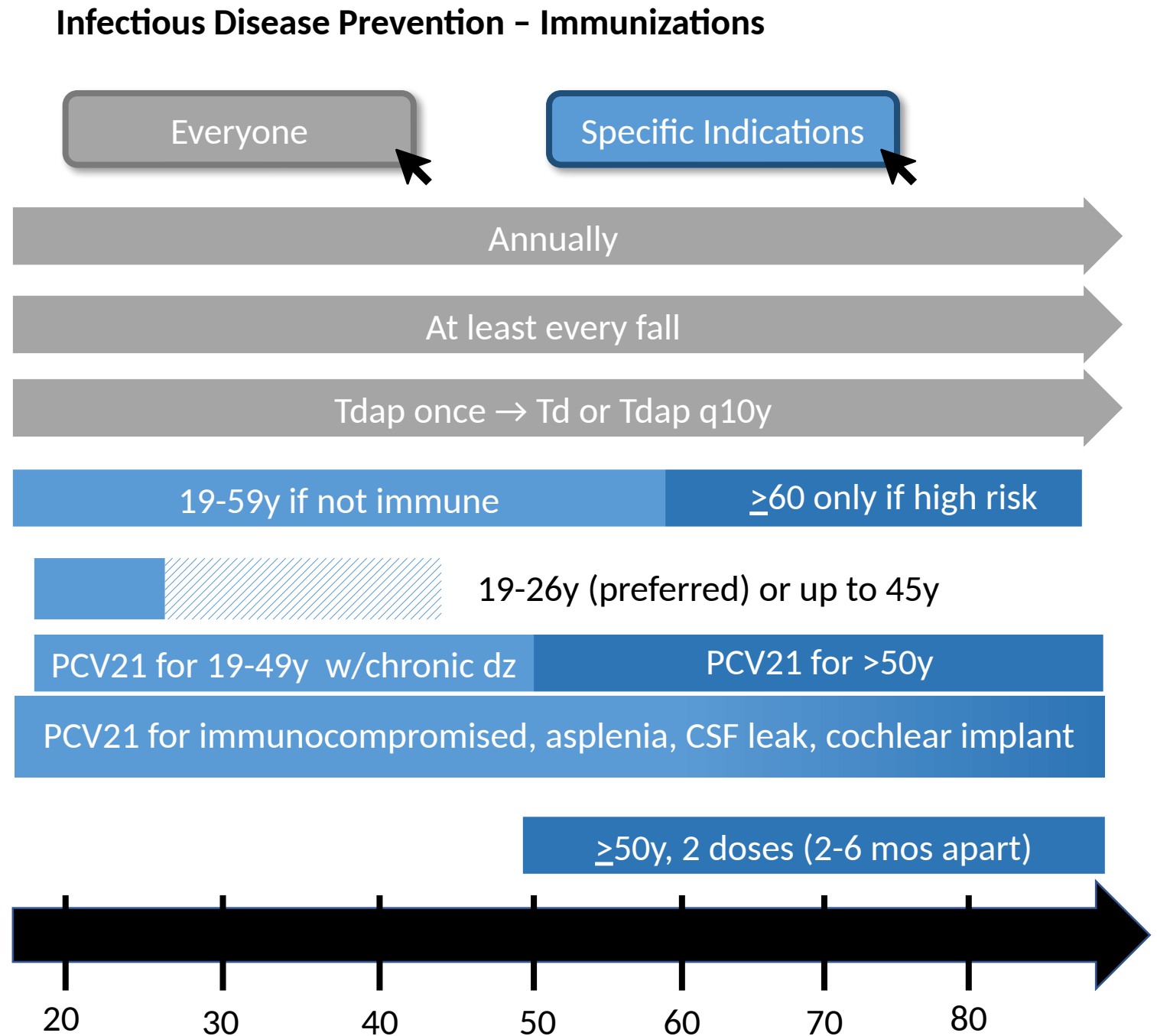
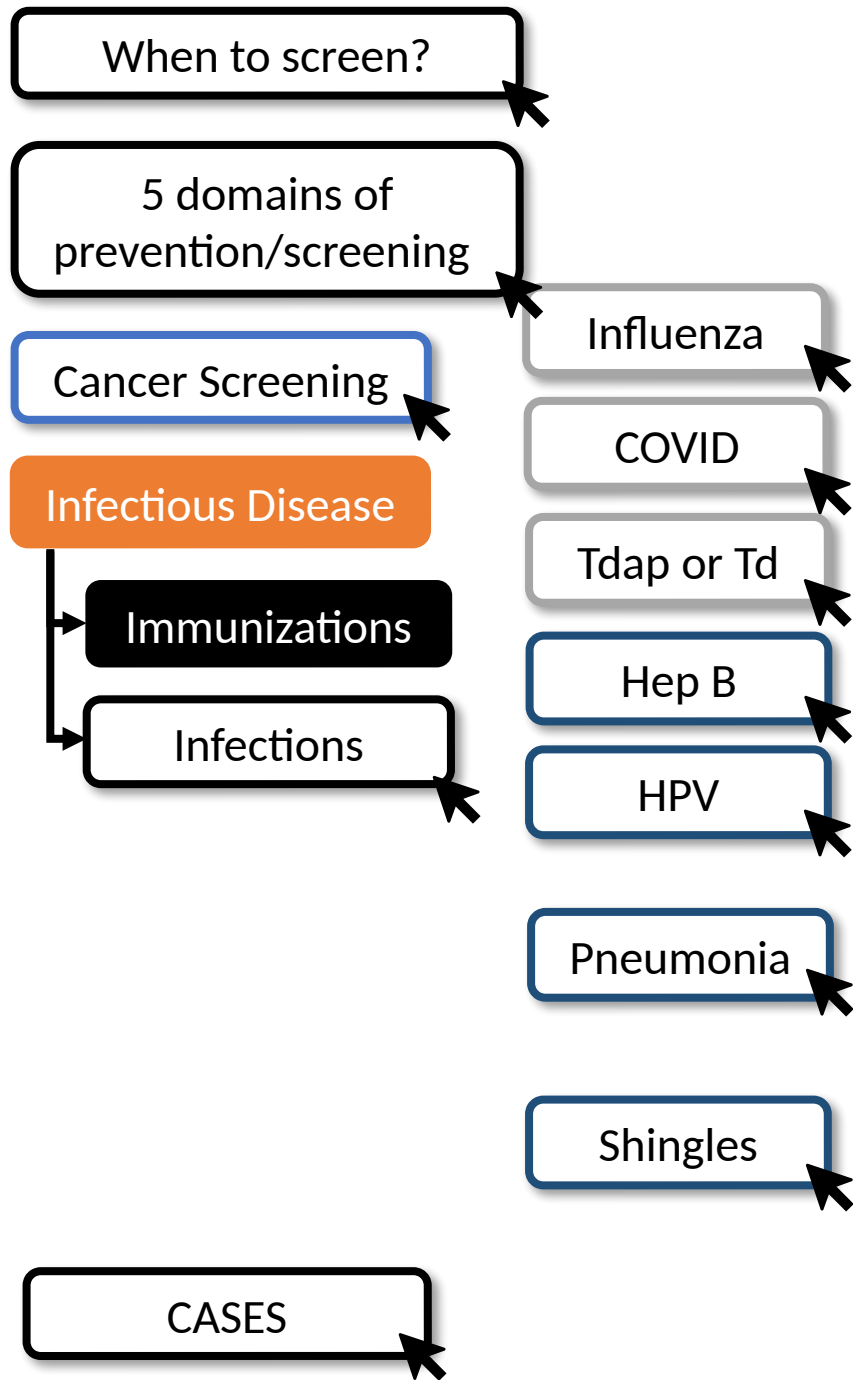
Modality and Interval?

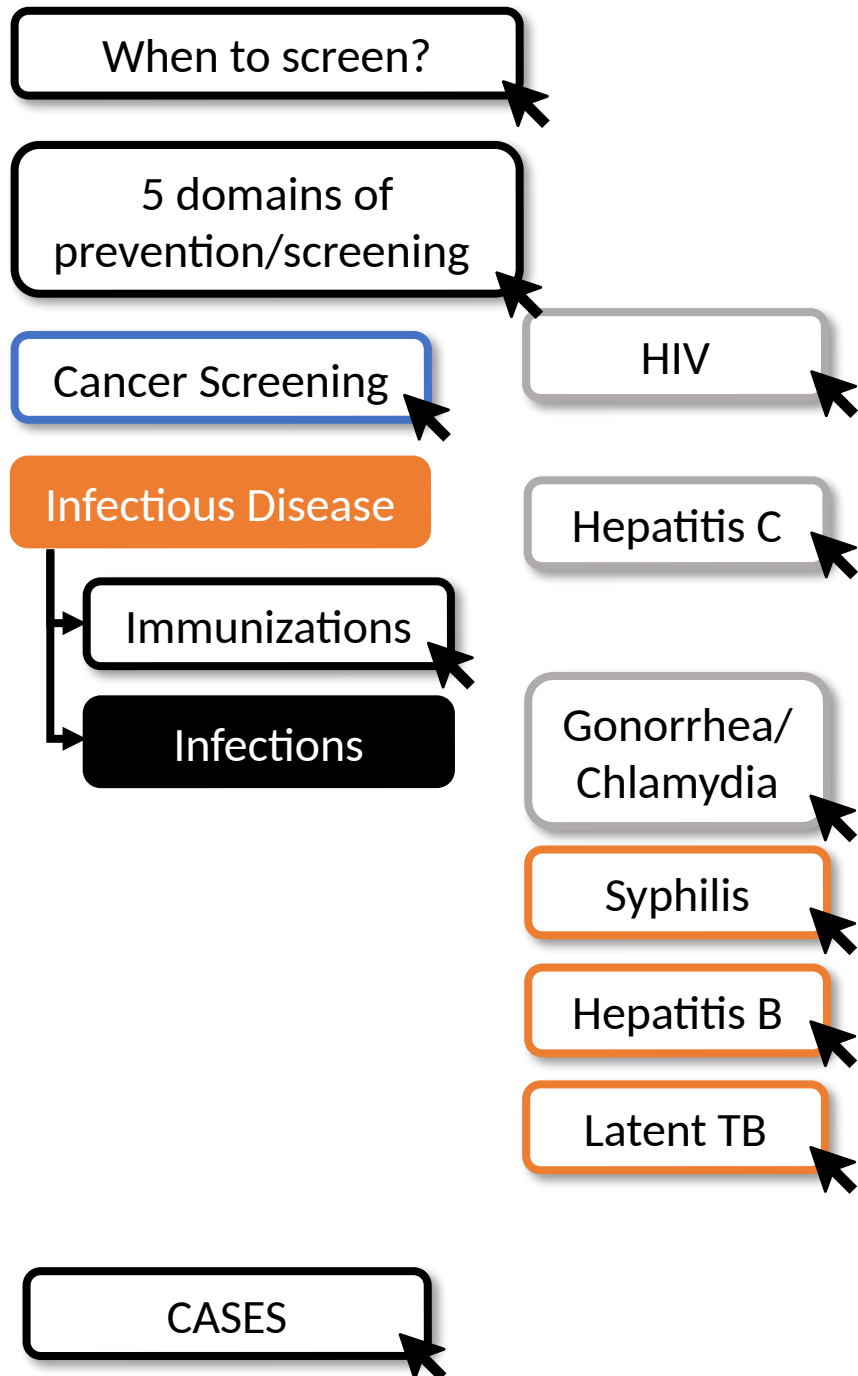
No screening or PSA
every 1-2y

Discontinuation?

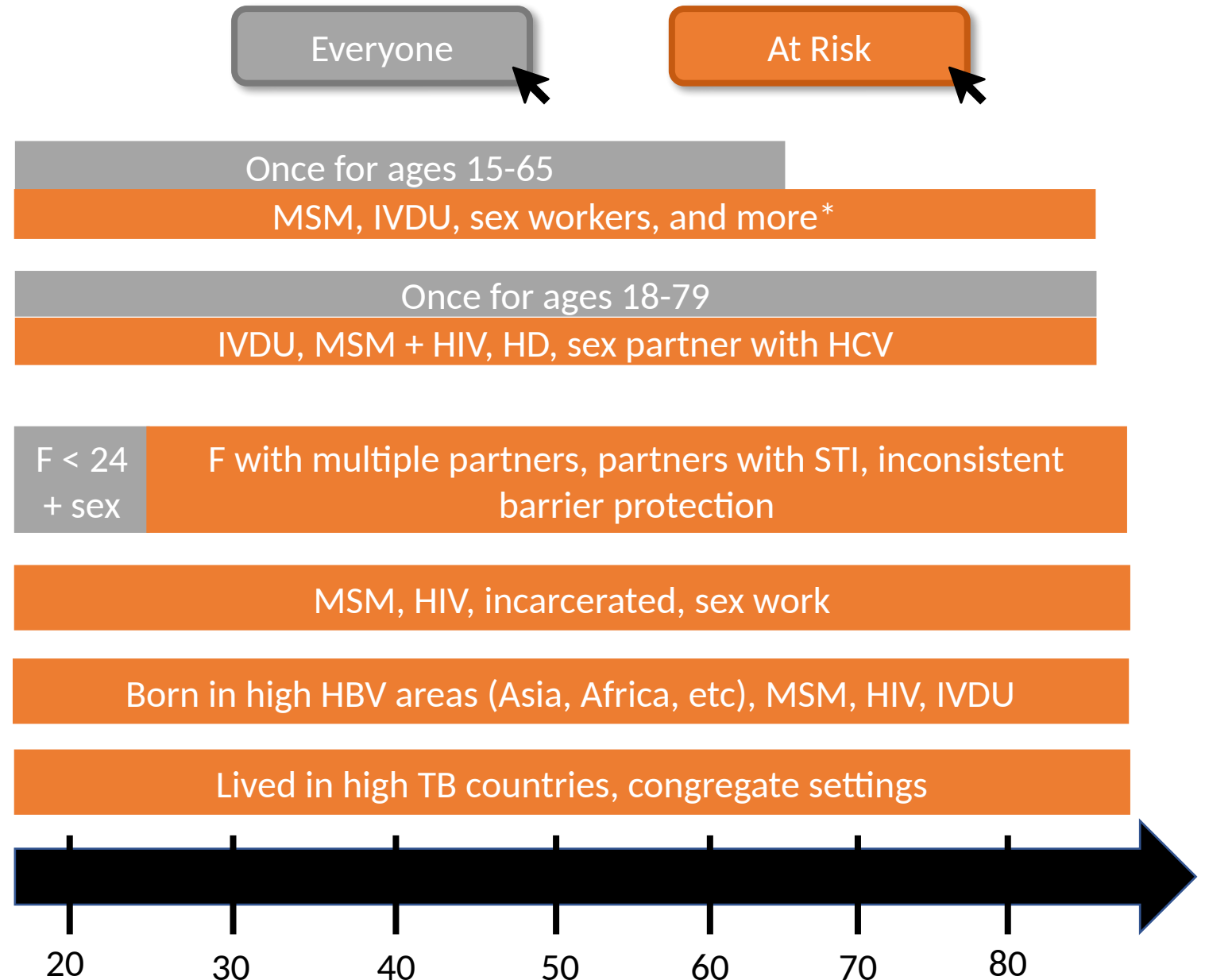
If >70y







Infectious Disease Prevention - Infection Screening



When to screen?

5 domains of screening

Cancer Screening

Infectious Disease

Immunizations

Infections

CASES

Case 1

54 yo W presents to establish care. She has a history of hypertension controlled on amlodipine 10 mg. She has smoked 1 ppd since she was 19 - tried quitting in the past but is not interested right now. Had a normal colonoscopy 4 years ago and a mammogram 2 years ago. She can't recall her last pap smear but has never had abnormal results. No family history of cancer. Sexually active with her husband.

VS: Ht 5'4", Wt 145lbs, HR 76, BP 132/75, RR 18, SpO2 94%

PE: Unremarkable cardiac and pulmonary exams.

1) Breast cancer, 2) Lung cancer, 3) Cervical cancer

- 1) Bilateral screening mammogram
- 2) LDCT scan after [shared decision making discussion](#)
- 3) Primary hrHPV test or Cytology + hrHPV co-test

Case 2

56 yo homeless M presents to establish care. He has not seen a provider for 8 years. He takes no medications and has no allergies. He has smoked ½ ppd for the past 20 years. Denies any illicit drug use and quit drinking alcohol 2 years ago. No family history of cancer and not currently sexually active.

VS: Ht 5'10", Wt 165lbs, HR 82, BP 138/80, RR 16, SpO2 95%

PE: Unremarkable cardiac and pulmonary exams.

1) Colon Cancer, 2) +/- Prostate Cancer

- 1) [FIT, gFOBT or FIT+DNA for colon cancer screening](#)
- 2) Discuss prostate cancer screening, likely recommend against it at this time

Case 3

32 yo G2P2 W presents for a physical. She has hypothyroidism on levothyroxine. She is sexually active with her boyfriend and has an IUD in place. Her last pap smear was 3 years ago when she was pregnant with her second child and it was normal. HPV testing was not done at that time. She does not know her family history of cancer.

Cervical cancer screening

Primary hrHPV or hrHPV + cytology co-test

[Repeat testing in 1 year](#)

2025 Update — USPSTF Cervical Cancer Screening

The deck's cancer-screening framework, breast (40–74 q2y, 2024), colon (45–79), lung (50–80/20py/15y), prostate (55–69 Grade C), PCV21, Hep B 19–59 and infection screening were all verified **CURRENT** (deck is the June-2025 build).

NEW — USPSTF finalized cervical cancer screening (2025), Grade A:

- Ages 21–29: cytology alone every 3 years (unchanged).
- Ages 30–65: high-risk HPV PRIMARY screening every 5 years is now the **PREFERRED** strategy.
- **FIRST-TIME** endorsement of **SELF-COLLECTED** (patient-collected) HPV samples — clinician- OR patient-collected accepted; aims to expand access.
- Alternatives at 30–65: cytology alone q3y, or HPV+cytology co-test q5y.

Caveat: ACS still starts at 25; ASCP/CETC raised concerns about equivalence and the 5-yr interval for self-collected samples. Confirm your institution's protocol.

Take-Home Points

Screening pays off only when the disease is common, has a detectable asymptomatic phase, early treatment beats late, and the test is accurate, cheap, and low-harm — benefits must outweigh harms/cost.

Five domains of adult prevention: Cancer, Infectious Disease, Cardiovascular, Psychosocial, Family/Safety. Individualize by age, sex, risk factors, comorbidities, and life expectancy.

Only 5 cancers have screening: breast (40–74 q2y), colon (45–79), cervical (21–65), lung (50–80, 20 pack-yr, quit <15y), prostate (55–69, shared decision, Grade C).

Immunizations 2024–25: PCV21 for adults ≥ 50 (and 19–49 with chronic disease); RSV vaccine for ≥ 75 (and 60–74 at higher risk) via shared decision-making; Hep B universal 19–59.

2025: USPSTF endorses self-collected HPV; primary HPV q5y preferred at 30–65. Use the USPSTF tool to keep guidelines straight.

Attribution & Optimization Notes

Adapted from TeachIM.org — “Adult Disease Prevention Part 1: Cancer Screening and Infectious Diseases” (Melton; Exec. Ed. Zhang; Sec. Ed. Brett; Univ. of Colorado / Univ. of Washington). Published Sept 2021, last reviewed June 2025.

Source URL: https://teachim.org/teaching_material/adult-disease-prevention-1/ | Deck: prevention-1-updated-2025.pptx (embed-verified)

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Optimized June 2026: original interactive board preserved (all 11 slides, animations & BioRender graphics intact). 1 surgical fix: slide 6 button label "USPTF" → "USPSTF" (single text run; animations preserved). 3 slides appended (this addendum).

Currency verified vs USPSTF (2024 breast; 2025 cervical self-collection final), ACIP 2024 (PCV21), and CDC RSV guidance. Companions: teaching script DOCX, cancer-screening learner handout PPTX, PCE summary PPTX.

Educational use only — clinician-reviewed before teaching; guidelines change frequently (check USPSTF/ACIP and local protocol).